

KuaiRand

An Unbiased Sequential Recommendation Dataset with Randomly Exposed Videos

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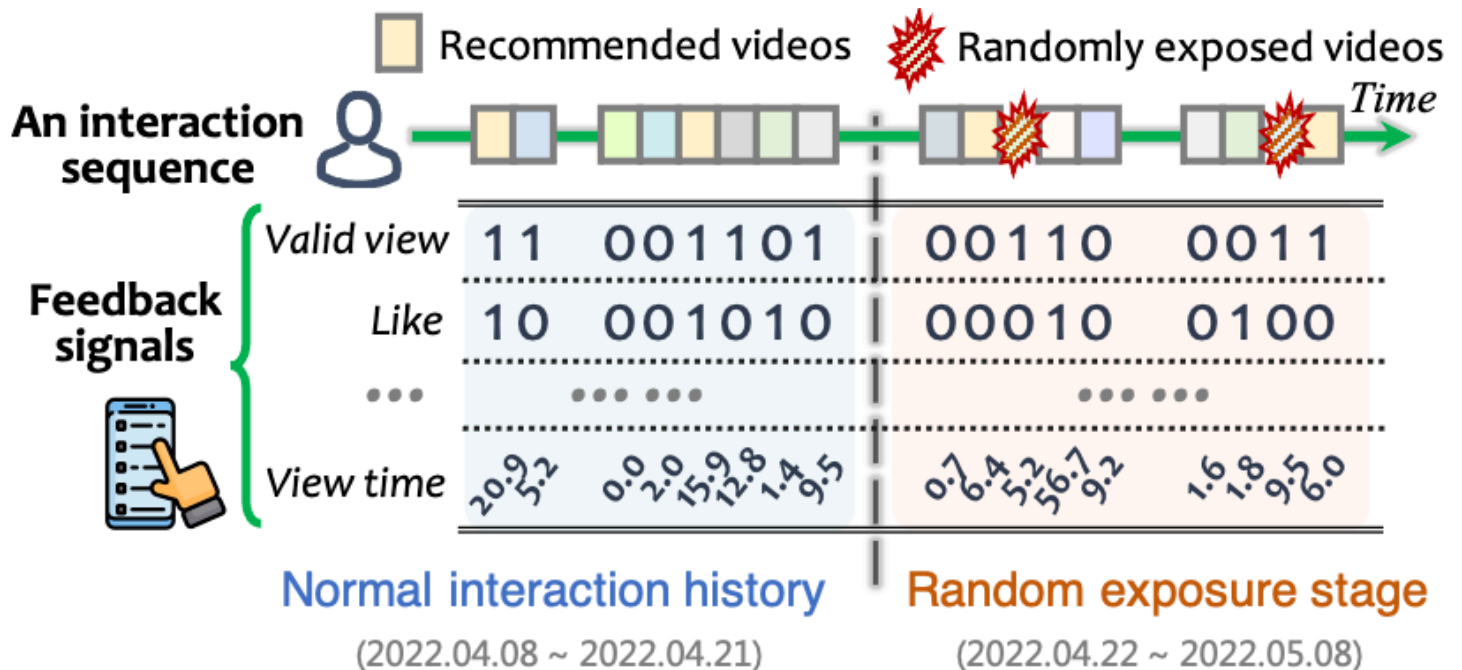
[45/3511808.3557624](#)

A sequential recommendation dataset collected from the popular video-sharing mobile app, [Kuaishou \(快手\)](#). It is the first dataset **with millions of intervened interactions of randomly exposed videos in standard recommendation feeds!**

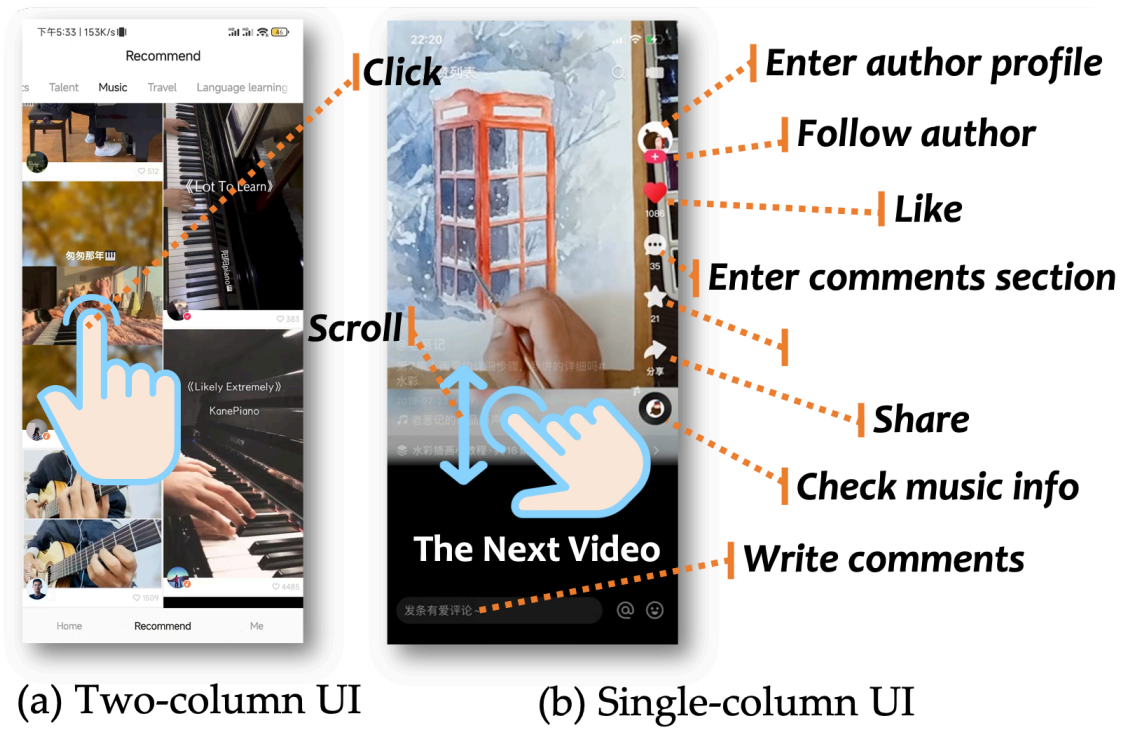
Other related **open-sourced** datasets are [KuaiRec](#) and [KuaiSAR](#).

Overview:

The following figure gives an example of the dataset. It illustrates a user interaction sequence along with the user's rich feedback signals.



These feedback signals are collected from the two main user interfaces (UI) in the Kuaishou APP shown as follows:



(a) Two-column UI

(b) Single-column UI

In the random exposure stage, each recommended video in the dataset has an equal probability of being replaced by a random video sampled from an item pool. About 0.37% Interactions are replaced in the final results.

Advantages:

Compared with other datasets with random exposure, KuaiRand has the following advantages:

- ✓ It is the first sequential recommendation dataset with millions of intervened interactions of randomly exposed items inserted in the standard recommendation feeds.
- ✓ It has the most comprehensive side information including explicit user IDs, interaction timestamps, and rich features for users and items.
- ✓ It has 15 policies with each catered for a special recommendation scenario in the Kuaishou App.
- ✓ We introduce 12 feedback signals (e.g., click, like, and view time) for each interaction to describe the user's comprehensive feedback.
- ✓ Each user has thousands of historical interactions on average.
- ✓ It has three versions to support various research directions in recommendation.

If you find it helpful, please cite our paper: [Paper Link](#) [PDF](#)

```
@inproceedings{gao2022kuairand,
  title = {KuaiRand: An Unbiased Sequential Recommendation Dataset with Randomly Intervened Interactions},
  author = {Gao, Chongming and Li, Shijun and Zhang, Yuan and Chen, Jiawei and Li, Jiayuan},
  url = {https://doi.org/10.1145/3511808.3557624},
  doi = {10.1145/3511808.3557624},
  booktitle = {Proceedings of the 31st ACM International Conference on Information and Knowledge Discovery},
  series = {CIKM '22},
```

```
location = {Atlanta, GA, USA},
numpages = {5},
year = {2022},
pages = {3953–3957}
}
```

Download the data:

KuaiRand has been shared at <https://zenodo.org/records/10439422>.

OPTION 1. Download via your browser:

You can download the dataset from the [zenodo site](#), or this [Chinese site](#).

OPTION 2: Download via the ‘wget’ command tool:

For the **KuaiRand-Pure** dataset:

```
wget https://zenodo.org/records/10439422/files/KuaiRand-Pure.tar.gz # (md5:082033.
tar -xzvf KuaiRand-Pure.tar.gz
```

For the **KuaiRand-1K** dataset:

```
wget https://zenodo.org/records/10439422/files/KuaiRand-1K.tar.gz # (md5:6b0b9c8.
tar -xzvf KuaiRand-1K.tar.gz
```

For the **KuaiRand-27K** dataset:

```
wget https://zenodo.org/records/10439422/files/KuaiRand-27K.tar.gz # (md5:3e3c799.
tar -xzvf KuaiRand-27K.tar.gz
```

Attention: We further provide two supplementary files that enrich the KuaiRand dataset with **video-level content-side semantic information**, namely `kuairand_video_categories.csv` and `kuairand_video_captions.csv`. Detailed descriptions and download links for these two files are provided [here](#).

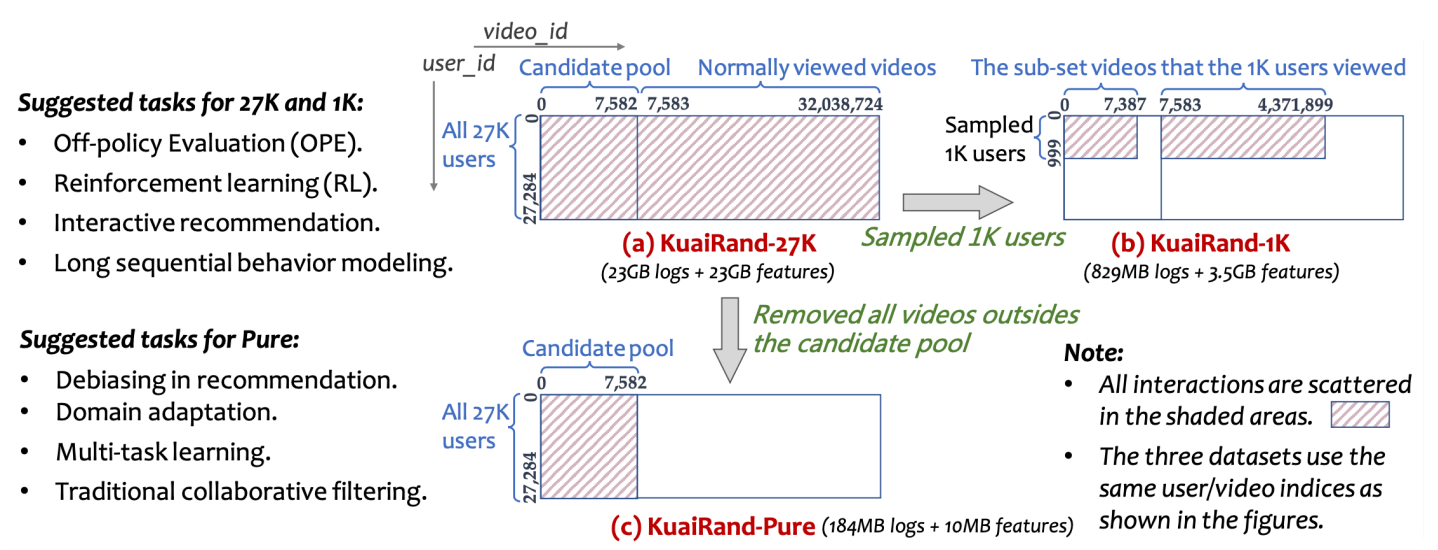
Three Versions and Suggestions:

We release three versions of KuaiRand for different uses:

- **KuaiRand-27K (23GB logs +23GB features):** the complete KuaiRand dataset that has over 27K users and 32 million videos.

- **KuaiRand-1K (829MB logs + 3.5GB features):** randomly sample 1,000 users from *KuaiRand-27K*, then remove all irrelevant videos. There are 4 million videos rest.
- **KuaiRand-Pure (184MB logs + 10MB features):** only keeps the logs for the 7583 videos in the candidate pool.

The *user_id* and *video_id* are re-indexed. A visualization of their ID spaces is shown as follows.



The basic statistics of the three versions are summarized as follows:

Dataset	Collection Policy	#Users	#Items	#Interactions	#User features	#Item features
KuaiRand-27K	15 policies	27,285	32,038,725	322,278,385	30	60
	Random policy	27,285	7,583	1,186,059	30	60
KuaiRand-1K	15 policies	1,000	4,369,953	11,713,045	30	60
	Random policy	1,000	7,388	43,028	30	60
KuaiRand-Pure	15 policies	27,285	7,551	1,436,609	30	60
	Random policy	27,285	7,583	1,186,059	30	60

Which version should I use?

- **Reasons to use *KuaiRand-27K* or *KuaiRand-1K*:**
 - Your research needs rigorous sequential logs, such as off-policy evaluation (OPE), Reinforcement learning (RL), or long sequential recommendation.
- **Reasons to use *KuaiRand-Pure*:**
 - The sequential information is not necessary for your research OR If you are OK with the incomplete sequential logs. For example, if you are studying debiasing in collaborative filtering models or multi-task modeling in recommendation.
 - If your model can only run with small-size data.

Data Descriptions

The file structure of the three datasets is listed as follows:

KuaiRand-27K (46GB)

```
KuaiRand-27K
├── data (46GB)
│   ├── log_random_4_22_to_5_08_27k.csv (83MB)
│   ├── log_standard_4_08_to_4_21_27k_part1.csv (4.8GB)
│   ├── log_standard_4_08_to_4_21_27k_part2.csv (4.8GB)
│   ├── log_standard_4_22_to_5_08_27k_part1.csv (6.6GB)
│   ├── log_standard_4_22_to_5_08_27k_part2.csv (6.6GB)
│   ├── user_features_27k.csv (3.4MB)
│   ├── video_features_basic_27k.csv (2.6GB)
│   ├── video_features_statistic_27k_part1.csv (6.7GB)
│   ├── video_features_statistic_27k_part2.csv (6.7GB)
│   └── video_features_statistic_27k_part3.csv (6.7GB)
└── load_data_27k.py
```

KuaiRand-1K (4.3GB)

```
KuaiRand-1K
├── data (4.3GB)
│   ├── log_random_4_22_to_5_08_1k.csv (2.9MB)
│   ├── log_standard_4_08_to_4_21_1k.csv (368MB)
│   ├── log_standard_4_22_to_5_08_1k.csv (481MB)
│   ├── user_features_1k.csv (132KB)
│   ├── video_features_basic_1k.csv (368MB)
│   └── video_features_statistic_1k.csv (3.1GB)
└── load_data_1k.py
```

KuaiRand-Pure (194MB)

```
KuaiRand-Pure
├── data (194MB)
│   ├── log_random_4_22_to_5_08_pure.csv (83MB)
│   ├── log_standard_4_08_to_4_21_pure.csv (80MB)
│   ├── log_standard_4_22_to_5_08_pure.csv (21MB)
│   ├── user_features_pure.csv (3.4MB)
│   ├── video_features_basic_pure.csv (612KB)
│   └── video_features_statistic_pure.csv (6.3MB)
└── load_data_pure.py
```

1 Description of the fields in `log_xxx.csv`

There are three log files:

- `log_random_4_22_to_5_08.csv` contains all interactions resulting from random intervention.
- `log_standard_4_22_to_5_08.csv` contains all interactions of standard recommendation.
- `log_standard_4_08_to_4_21.csv` contains all interactions of standard recommendation for the same users in the previous two weeks (2022.04.08 ~ 2022.04.21).

Field Name:	Description	Type	Example
user_id	The ID of the video.	int64	17387
video_id	The ID of the video.	int64	1123453
date	The date of this interaction	int64	20220421
hourmin	The time of this interaction (format: HHSS).	int64	400
time_ms	The timestamp of this interaction in milliseconds.	int64	1650485801301
is_click	A binary feedback signal. In the two-column UI, it indicates a click; In the single-column UI, it means <i>valid_play</i> , which equals 1 when: <code>play_time_ms >= duration_ms</code> if <code>duration_ms <= 7,000 ms</code> , or <code>play_time_ms > 7,000 ms</code> if <code>duration_ms > 7,000 ms</code> .	int64	1

Field Name:	Description	Type	Example
is_like	A binary feedback signal indicating the user hit the <code>like</code> button.	int64	0
is_follow	A binary feedback signal indicating the user hit the <code>follow the author</code> button.	int64	0
is_comment	A binary feedback signal indicating the user wrote a comment in the comments section of this video	int64	0
is_forward	A binary feedback signal indicating the user forwarded this video.	int64	0
is_hate	A binary feedback signal indicating the user hated this video.	int64	0
long_view	A binary feedback signal. It equals 1 when: <code>play_time_ms >= duration_ms</code> if <code>duration_ms <= 18,000 ms</code> , or <code>play_time_ms >= 18,000 ms</code> if <code>duration_ms > 18,000 ms</code> .	int64	1
play_time_ms	The user's view time in milliseconds.	int64	151024
duration_ms	The video's duration time in milliseconds.	int64	104400
profile_stay_time	The time that the user stayed in this author's profile.	int64	0
comment_stay_time	The time that the user stayed in the comments section of this video	int64	0
is_profile_enter	A binary feedback signal indicating the user enters the author profile	int64	0
is_rand	A binary signal indicating if this video is generated by the random intervention (i.e., a random exposed video).	int64	0

Field Name:	Description	Type	Example
tab	indicating the scenario of this interaction, e.g., in the recommendation page or the main page of the App. In the range of [0,14].	int64	1

2 Descriptions of the fields in `user_features.csv`

Field Name:	Description	Type	Example
user_id	The ID of the user.	int64	25621
user_active_degree	In the set of {'high_active', 'full_active', 'middle_active', 'UNKNOWN'}.	str	"full_active"
is_lowactive_period	Is this user in its low active period	int64	0
is_live_streamer	Is this user a live streamer?	int64	0
is_video_author	Has this user uploaded any video?	int64	1
follow_user_num	The number of users that this user follows.	int64	5
follow_user_num_range	The range of the number of users that this user follows. In the set of {'0', '(0,10]', '(10,50]', '(100,150]', '(150,250]', '(250,500]', '(50,100]', '500+'}	str	"(0,10]"
fans_user_num	The number of the fans of this user.	int64	312
fans_user_num_range	The range of the number of fans of this user. In the set of {'0', '[1,10)', '[10,100)', '[100,1k)', '[1k,5k)', '[5k,1w)', '[1w,10w)'}	str	"[100,1k)"
friend_user_num	The number of friends that this user has.	int64	0

Field Name:	Description	Type	Example
friend_user_num_range	The range of the number of friends that this user has. In the set of {'0', '[1,5)', '[5,30)', '[30,60)', '[60,120)', '[120,250)', '250+'}	str	"0"
register_days	The days since this user has registered.	int64	3624
register_days_range	The range of the registered days. In the set of {'15-30', '31-60', '61-90', '91-180', '181-365', '366-730', '730+'}.	str	"730+"
onehot_feat0	An encrypted feature of the user. Each value indicates the position of "1" in the one-hot vector. Range: {0,1}	int64	1
onehot_feat1	An encrypted feature. Range: {0, 1, ..., 6}	int64	2
onehot_feat2	An encrypted feature. Range: {0, 1, ..., 49}	int64	2
onehot_feat3	An encrypted feature. Range: {0, 1, ..., 1470}	int64	1153
onehot_feat4	An encrypted feature. Range: {0, 1, ..., 14}	int64	4
onehot_feat5	An encrypted feature. Range: {0, 1, ..., 33}	int64	0
onehot_feat6	An encrypted feature. Range: {0, 1, 2}	int64	0
onehot_feat7	An encrypted feature. Range: {0, 1, ..., 117}	int64	31
onehot_feat8	An encrypted feature. Range: {0, 1, ..., 453}	int64	354
onehot_feat9	An encrypted feature. Range: {0, 1, ..., 6}	int64	3
onehot_feat10	An encrypted feature. Range: {0, 1, ..., 4}	int64	3

Field Name:	Description	Type	Example
onehot_feat11	An encrypted feature. Range: {0, 1, ..., 4}	int64	2
onehot_feat12	An encrypted feature. Range: {0, 1}	int64	1
onehot_feat13	An encrypted feature. Range: {0, 1}	int64	0
onehot_feat14	An encrypted feature. Range: {0, 1}	int64	0
onehot_feat15	An encrypted feature. Range: {0, 1}	int64	0
onehot_feat16	An encrypted feature. Range: {0, 1}	int64	0
onehot_feat17	An encrypted feature. Range: {0, 1}	int64	0

3 Descriptions of the fields in `video_features_basic.csv`

Field Name:	Description	Type	Example
video_id	The ID of the video.	int64	3784
author_id	The ID of the author of this video. In the range of [0, 8839734]	int64	441
video_type	Type of this video (NORMAL or AD).	str	"NORMAL"
upload_dt	Upload date of this video.	str	"2020-07-08"
upload_type	The upload type of this video.	str	"ShortImport"
visible_status	The visible state of this video on the APP now.	int	1
video_duration	The time duration of this duration (in milliseconds).	Int64	17200.0
server_width	The width of this video on the server.	int64	720
server_height	The height of this video on the server.	int64	1280
music_id	Background music ID of this video.	int64	989206467
music_type	Background music type of this video.	int64	4

Field Name:	Description	Type	Example
tag	A list of key categories (labels) of this video.	str	"12,65"

4 Descriptions of the fields in video_features_statistic.csv

!! Different from the basic features, the statistical features are the average statistics of the video each day over one month. For example, in the following table, video 9288071 has 66 counts over this one month (a video can have multiple counts each day on different scenarios, e.g., on April 8, `show_cnt` =80 on the main page and `show_cnt` =65 on the recommendation page of the App)

Field Name:	Description	Type	Example
video_id	The ID of the video.	int64	9288071
counts	The number of statistics.	int64	66
show_cnt	The number of shows of this video (averaged on each day and each scenario over one month. This applies to all the following fields)	float64	75.212
show_user_num	The number of users who received the recommendation of this video.	float64	66.985
play_cnt	The number of plays.	float64	9.409
play_user_num	The number of users who play this video.	float64	8.121
play_duration	The total time duration of playing this video (in milliseconds).	float64	93700.333
complete_play_cnt	The number of complete plays. <i>complete play</i> : finishing playing the whole video, i.e., <code> #(play_duration >= video_duration) .</code>	float64	0.182
complete_play_user_num	The number of users who perform the <i>complete play</i> .	float64	0.182
valid_play_cnt	<i>valid play</i> : <code>play_duration >= video_duration</code> if <code>video_duration <= 7s</code> , or	float64	3.545

Field Name:	Description	Type	Example
	<code>play_duration > 7</code> if <code>video_duration > 7s</code> .		
valid_play_user_num	The number of users who perform the <i>complete play</i> .	float64	3.136
long_time_play_cnt	<i>long time play</i> : <code>play_duration >= video_duration</code> if <code>video_duration <= 18s</code> , or <code>play_duration >= 18</code> if <code>video_duration > 18s</code> .	float64	1.909
long_time_play_user_num	The number of users who perform the <i>long time play</i> .	float64	1.848
short_time_play_cnt	<i>short time play</i> : <code>play_duration < min(3s, video_duration)</code> .	float64	5.015
short_time_play_user_num	The number of users who perform the <i>short time play</i> .	float64	4.545
play_progress	The average video playing ratio (<code>=play_duration/video_duration</code>)	float64	0.016
comment_stay_duration	Total time of staying in the comments section	float64	2302.712
like_cnt	Total likes	float64	0.303
like_user_num	The number of users who hit the “like” button.	float64	0.303
click_like_cnt	The number of the “like” resulted from double click	float64	0.030
double_click_cnt	The number of users who double-click the video.	float64	0.273
cancel_like_cnt	The number of likes that are canceled by users.	float64	0.485
cancel_like_user_num	The number of users who cancel their likes.	float64	0.485

Field Name:	Description	Type	Example
comment_cnt	The number of comments within this day.	float64	0.015
comment_user_num	The number of users who comment on this video.	float64	0.015
direct_comment_cnt	The number of direct comments (depth=1).	float64	0.015
reply_comment_cnt	The number of reply comments (depth>1).	float64	0.000
delete_comment_cnt	The number of deleted comments.	float64	0.015
delete_comment_user_num	The number of users who delete their comments.	float64	0.015
comment_like_cnt	The number of comment likes.	float64	0.000
comment_like_user_num	The number of users who like the comments.	float64	0.000
follow_cnt	The number of increased follows from this video.	float64	0.000
follow_user_num	The number of users who follow the author of this video due to this video.	float64	0.000
cancel_follow_cnt	The number of decreased follows from this video.	float64	0.000
cancel_follow_user_num	The number of users who cancel their following of the author of this video due to this video.	float64	0.000
share_cnt	The times of successfully sharing this video.	float64	0.000
share_user_num	The number of users who succeed to share this video.	float64	0.000
download_cnt	The times of downloading this video.	float64	0.030

Field Name:	Description	Type	Example
download_user_num	The number of users who download this video.	float64	0.030
report_cnt	The times of reporting this video.	float64	0.000
report_user_num	The number of users who report this video.	float64	0.000
reduce_similar_cnt	The times of reducing similar content of this video.	float64	0.015
reduce_similar_user_num	The number of users who choose to reduce similar content of this video.	float64	0.015
collect_cnt	The times of adding this video to favorite videos.	float64	0.061
collect_user_num	The number of users who add this video to their favorite videos.	float64	0.061
cancel_collect_cnt	The times of removing this video from favorite videos.	float64	0.091
cancel_collect_user_num	The number of users who remove this video from their favorite videos	float64	0.091
direct_comment_user_num	The number of users who write comments directly under this video (level=1).	float64	0.015
reply_comment_user_num	The number of users who reply the existing comments (level>1).	float64	0.000
share_all_cnt	The times of sharing this video (no need to be successful).	float64	0.015
share_all_user_num	The number of users who share this video (no need to be successful).	float64	0.015
outside_share_all_cnt	The times of sharing this video outside Kuaishou App.	float64	0.000

Additional Video-Level Metadata (Supplementary Files)

We further provide two supplementary files that enrich KuaiRand with **content-side semantic information** at the video level. These files can be joined with the main dataset using `final_video_id`.

Two files: `kuairand_video_captions.csv` and `kuairand_video_categories.csv` are available via <https://zenodo.org/records/18159199>

or you can just download via the **‘wget’ command tool**:

1. `kuairand_video_captions.csv` :

```
wget https://zenodo.org/records/18159199/files/kuairand_video_captions.csv # (md5
```

1. `kuairand_video_categories.csv` :

```
wget https://zenodo.org/records/18159199/files/kuairand_video_categories.csv # (l
```

 Descriptions of the fields in `kuairand_video_captions.csv`

Field Name:	Description	Type	Example
<code>final_video_id</code>	The re-indexed ID of the video, consistent with KuaiRand.	<code>int64</code>	6
<code>caption</code>	The main caption text of the video, including hashtags and mentions.	<code>str</code>	#王者荣耀妲己秀 #游戏凡星计划 刺客法爆流妲己教程在下方小黄车哟!
<code>show_cover_text</code>	Text displayed on the video cover (may be empty).	<code>str</code>	妲己打亚瑟 就怕他六分投
<code>duration</code>	Video duration in milliseconds.	<code>float64</code>	114500.0

This file provides **textual and semantic content information** of videos, which can be used for multimodal recommendation, text-aware modeling, and content understanding tasks.

 Descriptions of the fields in `kuairand_video_categories.csv`

Field Name:	Description	Type	Example
final_video_id	The re-indexed ID of the video, consistent with KuaiRand.	int64	3
first_level_category_id	ID of the first-level (top) category.	float64	7.0
first_level_category_name	Name of the first-level category.	str	运动
first_level_category_prob	Prediction probability of the first-level category.	float64	0.9996
second_level_category_id	ID of the second-level category.	float64	126.0
second_level_category_name	Name of the second-level category.	str	球类运动
second_level_category_prob	Prediction probability of the second-level category.	float64	1.0
third_level_category_id	ID of the third-level category (if available).	float64	1056.0
third_level_category_name	Name of the third-level category.	str	篮球
third_level_category_prob	Prediction probability of the third-level category.	float64	1.0
fourth_level_category_id	ID of the fourth-level category (if available).	float64	-124.0
fourth_level_category_name	Name of the fourth-level category (UNKNOWN if unavailable).	str	UNKNOWN
fourth_level_category_prob	Prediction probability of the fourth-level category.	float64	0.0

The category system is **hierarchical (up to four levels)**.

If a category level is unavailable, its ID is set to **-124** and the name to **UNKNOWN** .

This file enables research on **category-aware recommendation, hierarchical classification,** and **content taxonomy modeling**.

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